

Collin Quirke

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OBJECTIVE

Mechanical Engineering student specialized in aerospace propulsion and solid rocket motor design, with experience in simulation (MATLAB, OpenMotor, RPA) and hardware development. Seeking hands on engineering roles propulsion, testing, or aerospace development.

EDUCATION

Bachelor of Science in Mechanical Engineering, Liberty University, Lynchburg, VA

May 2028

- GPA: 3.5/4.0

TECHNICAL SKILLS

Engineering & Analysis

- MATLAB (Advanced), OpenMotor, ANSYS (Basic)

CAD & Design

- SolidWorks, Inventor, Onshape, AutoCAD

Programming

- MATLAB, C++, HTML, CSS

Manufacturing & Hardware

- Lathe Operation, Laser Cutting, 3D Printing

CERTIFICATIONS

Tripoli L2

Anticipated May 2026

OnShape

May 2025

PROJECTS

Solid Rocket Motor Design Tool (MATLAB)

- Developed a MATLAB-based tool to automate solid rocket motor (SRM) design; including casing, nozzle, and grain geometry generation, stress analysis, and performance prediction
- Implemented user driven inputs to model geometry, structural loads, and thrust characteristics
- Reduced design iteration time by ~80%, enabling rapid evaluation of multiple motor configurations

High Power Rocket (Tripoli L2 Certification)

- Designed and built a high power rocket system to meet Tripoli Rocketry Association Level 2 certification requirements
- Selected and integrated H-class propulsion system, ensuring structural integrity and flight stability for the thrust profile
- Performed flight stability analysis (CG/CP margin) and validated safe recovery system deployment
- Conducted launch and post-flight evaluation, verifying system performance and recovery reliability

RELEVANT EXPERIENCE

Senior Casing Engineer, Liberty University Rocketry Team, Lynchburg, VA

September 2024-Current

- Designed and supported manufacturing of inaugural experimental solid rocket motor (SRM) operating at 600 psi chamber pressure
- Led development of a 38mm H-class motor using RXN 71V propellant, ensuring structural integrity and reliable performance
- Designed boat tail structure and coordinated with AeroStructures team to integrate with fin can assembly
- Improved aerodynamic efficiency while meeting budget constraints and achieving apogee within 2% of target
- Developed MATLAB scripts for SRM optimization and structural analysis, improving design accuracy and iteration speed
- Conducted testing and post analysis of motor performance, comparing predicted vs. actual thrust and burn characteristics

Intern, NASA, VASTS Program, Remote

November 2023-April 2024

- Performed orbital and deorbit trajectory analysis for conceptual space missions
- Estimated payload capacity and Δv requirements for mission feasibility studies
- Produced 6 technical reports and 2 engineering drawings communicating system level designs
- Collaborated in remote engineering team environment to develop and refine aerospace concepts

Team Lead, Chick-Fil-A, Winchester, VA

March 2022-November 2023

- Led team operations in high volume, fast paced environment, maintaining efficiency and service quality
- Developed strong communication, leadership, and problem skills under pressure